Expressions and operator order

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

Before you start

What is the difference between 8 + 2 and "8" + "2"?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Evaluate numeric expressions using Python’s grouping rules. Try the recall before opening notes.

Your learning target

I can evaluate numeric **expression**s using Python’s grouping rules.

I can use **parentheses** to express the intended calculation.

I can distinguish ordinary division, whole groups and a **remainder**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Evaluate numeric **expression**s using Python's grouping rules

Use **parentheses** to express the intended calculation

Distinguish ordinary division, whole groups and a **remainder**

Retrieve what you already know

> What is the difference between 8 + 2 and "8" + "2"?

My starting explanation

This week's end goal

Trace **expression**s and **operator** order and explain the changed program before running it.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Expressions and operator order case. Separate observations, given facts and assumptions before choosing an explanation.

Three packs each contain eight counters, with two more loose counters. First count the packed counters, then add the loose ones. A program needs that structure, not just the right numbers.

This week you will predict **expression**s, make grouping visible and check division against a physical interpretation.

Use numeric values in a fresh Python 3 script called **expression**s.py. Keep expected and observed results separate as before. Multiplication and division group more tightly than addition and subtraction.

Parentheses let you state a different grouping. A correct calculation still depends on correct quantities and units.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Expressions and operator order. Connect each labelled input, deciding step and result.

Two expressions with different meanings

EXPRESSION	FIRST CALCULATION	FINAL VALUE
3 * 8 + 2	3 * 8 is 24	24 + 2 is 26
3 * (8 + 2)	8 + 2 is 10	3 * 10 is 30
(3 * 8) + 2	3 * 8 is 24	24 + 2 is 26

Adding two inside every pack differs from adding two loose counters once.

```
packs = 3
per_pack = 8
loose = 2
total = packs * per_pack + loose
print(total)
```

The script displays 26. Replacing the total **expression** with packs * (per_pack + loose) displays 30, because it adds two to every pack. **Parentheses** are not decoration: they can change the represented situation.

Write a brief phrase for each group before running.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Expressions and operator order. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Replace the final multiplier 5 with 4 in both **expression**s.

Predicted result

Prediction: The first result is 11; the grouped result is 20.

Why the result follows

Reason: 3+2*4=3+8; (3+2)*4=5*4.

Words that unlock the idea

> Use the vocabulary of Expressions and operator order to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Expression	Code evaluated to produce a value
Operator	A symbol such as + or * that specifies an operation
Precedence	Rules that determine how an expression is grouped
Parentheses	Symbols that make a grouping explicit
Remainder	The amount left after forming whole groups

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Expressions and operator order, predict a step, then compare with its explanation.

```
print(20 - 6 - 2)
print(20 - (6 - 2))
print(12 / 3 * 2)
```

Worked explanation · Expressions and operator order

The first line groups subtraction from left to right: (20 - 6) - 2 gives 12. The second gives 16. Multiplication and division share a **precedence** level, so 12 / 3 * 2 groups as (12 / 3) * 2 and displays 8.0, not 2.0.

Ordinary / division produces a float, even when the result is mathematically whole.

Explain why 18 - 4 * 3 gives 6 while (18 - 4) * 3 gives 42. Which operation is grouped first in each expression?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Expressions and operator order

Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

The facts we will use

Supplied facts

Python 3: print(3+2*5); print((3+2)*5). **Input**s are dimensionless for this arithmetic exercise.

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Expressions and operator order

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Use precedence

In 3+2*5, multiplication happens first.

2. Compute first result

2*5 is 10; adding 3 gives 13.

3. Compute grouped result

Parentheses give 3+2=5 first; multiplying by 5 gives 25.

CHANGE ONE THING

Replace the final multiplier 5 with 4 in both expressions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Use precedence	In 3+2*5, multiplication happens first.
2. Compute first result	2*5 is 10; adding 3 gives 13.
3. Compute grouped result	Parentheses give 3+2=5 first; multiplying by 5 gives 25.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

What this example does not establish

Keep the stated limit

Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

Keep the stated limit

Read the labelled picture: Trace the operation order in both **expression**s. Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

Keep the stated limit

Change one thing: Replace the final multiplier 5 with 4 in both **expression**s. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Expressions and operator order claim and explain a repair using the supplied evidence.

```
counters = 17
group_size = 5
print(counters / group_size)
print(counters // group_size)
print(counters % group_size)
```

Check the claim against the facts

The **output**s are 3.4, 3 and 2. With nonnegative whole counters and a positive group size, // gives the number of complete groups and % gives the **remainder**: 17 = 3 * 5 + 2.

Ordinary division answers a different question about a ratio; 3.4 complete five-counter groups cannot be built without splitting counters.

Check the claim against the facts

For nonnegative whole **input**s, the **remainder** is at least zero and smaller than the group size. A group size of zero is invalid and division by zero raises an error.

These whole-group interpretations assume nonnegative counts and positive group sizes; do not generalise them to negative quantities without examining Python's floor-division rule.

Locate the earliest unsupported step

For 17 counters in groups of 5, explain /, // and %. Show a multiplication-and-addition check for the complete groups and **remainder**.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Expressions and operator order with support, then explain the deciding step in your own words.

Write an **expression** for three packs of eight counters plus two loose counters. Explain why adding two within the **parentheses** for every pack describes a different situation.

Explain why 18 - 4 * 3 gives 6 while (18 - 4) * 3 gives 42. Which operation is grouped first in each expression?

Explain your prediction to a partner before comparing results.

Practise with less help: Predict then test expressions

Use workbook W2 for the connected example "**Expression**s and **operator** order". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Expression**s and **operator** order. Proposed explanation: "All operations are read left to right regardless of **precedence**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Trace the operation order in both **expression**s. Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Expressions and operator order. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

```
packs = 3
per_pack = 8
loose = 2
total = packs * per_pack + loose
print(total)
```

```
print(20 - 6 - 2)
print(20 - (6 - 2))
```

```
print(12 / 3 * 2)
```

```
counters = 17
group_size = 5
print(counters / group_size)
print(counters // group_size)
print(counters % group_size)
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Expressions and operator order. Explain why each step is justified.

Explain → apply → challenge

Write an **expression** for three packs of eight counters plus two loose counters. Explain why adding two within the **parentheses** for every pack describes a different situation.

For 17 counters in groups of 5, explain /, // and %. Show a multiplication-and-addition check for the complete groups and **remainder**.

Explain why a group size of zero must be excluded. State the assumptions that let complete groups and a **remainder** describe the packing task.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Python readiness portfolio. Keep an annotated decision and the evidence supporting it.

Add a labelled **expression**s and **operator** order **evidence** sheet to Python readiness portfolio. Show how you can evaluate numeric **expression**s using python's grouping rules.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	

Project evidence	Record
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Keep the **expression**, substituted numbers, grouping, units and actual result. A bare number does not show why it answers the question.

If an edit changes the grouping, preserve the old result and rerun a case where the **expression**s differ. Week 4 will turn comparisons into Boolean conditions.

> **A team has 23 whole tiles and puts 4 in each complete tray. Which operations find complete trays and leftover tiles? Predict both values and verify them by reconstructing 23. Explain why / alone does not answer both questions.**

Explain why a group size of zero must be excluded. State the assumptions that let complete groups and a **remainder** describe the packing task.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Expressions and operator order case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Two expressions with different meanings

EXPRESSION	FIRST CALCULATION	FINAL VALUE
3 * 8 + 2	3 * 8 is 24	24 + 2 is 26
3 * (8 + 2)	8 + 2 is 10	3 * 10 is 30
(3 * 8) + 2	3 * 8 is 24	24 + 2 is 26

Adding two inside every pack differs from adding two loose counters once.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Group an expression deliberately

Predict each line. Explain which calculation groups addition before division. What would (17+5)%5 return, and how do /, // and % differ for the supplied positive integers?

```
print((17 + 5) // 5)
print(17 + 5 // 5)
print(17 / 5, 17 // 5, 17 % 5)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Four trays hold six tiles each and five tiles are loose. Write and evaluate the total **expression**. Then pack the total into complete groups of four and find the **remainder**. Explain why 4 * (6 + 5) is wrong for this story.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Expressions and operator order, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: **Parentheses** change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

By the end, I can	How I will show it
Evaluate numeric expressions using Python's grouping rules	Explain the deciding step, show a trace or test, and state one limit.
Use parentheses to express the intended calculation	Explain the deciding step, show a trace or test, and state one limit.
Distinguish ordinary division, whole groups and a remainder	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Expressions and operator order without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Evaluate numeric **expression**s using Python's grouping rules

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.

1. Compute grouped result

2. Use **precedence**

3. Compute first result

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compute grouped result **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. **Parentheses** explicitly group an operation.

Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace the final multiplier 5 with 4 in both **expression**s. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The first result is 11; the grouped result is 20.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The first result is 11; the grouped result is 20. 3+2*4=3+8; (3+2)*4=5*4.

Repair a missing building block

Review Values variables and types (python-bridge/2). Goal: Distinguish integer, floating-point, text and Boolean values. Short practice: Predict both print **output**s and explain why they differ.

Does count+tag work, and what happens to each valid **output** after count becomes ten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

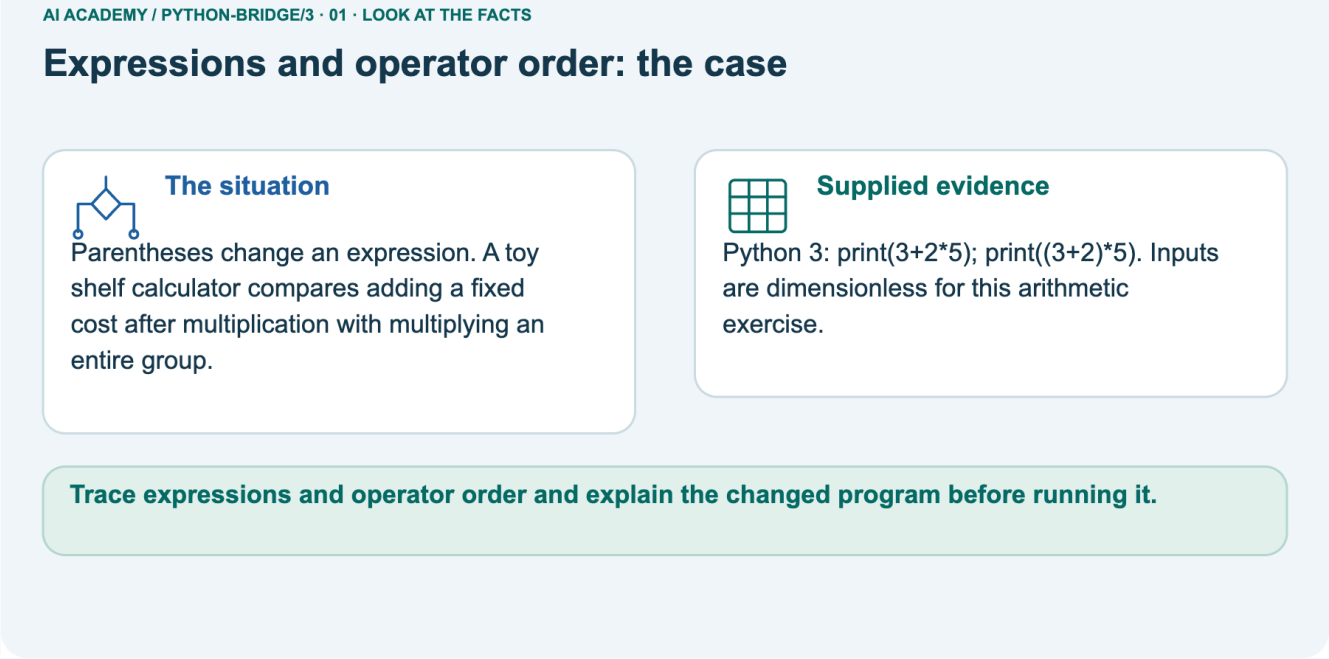
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Expressions and operator order graphic using its labels, deciding evidence and relevant limitation.

01. Expressions and operator order: the case



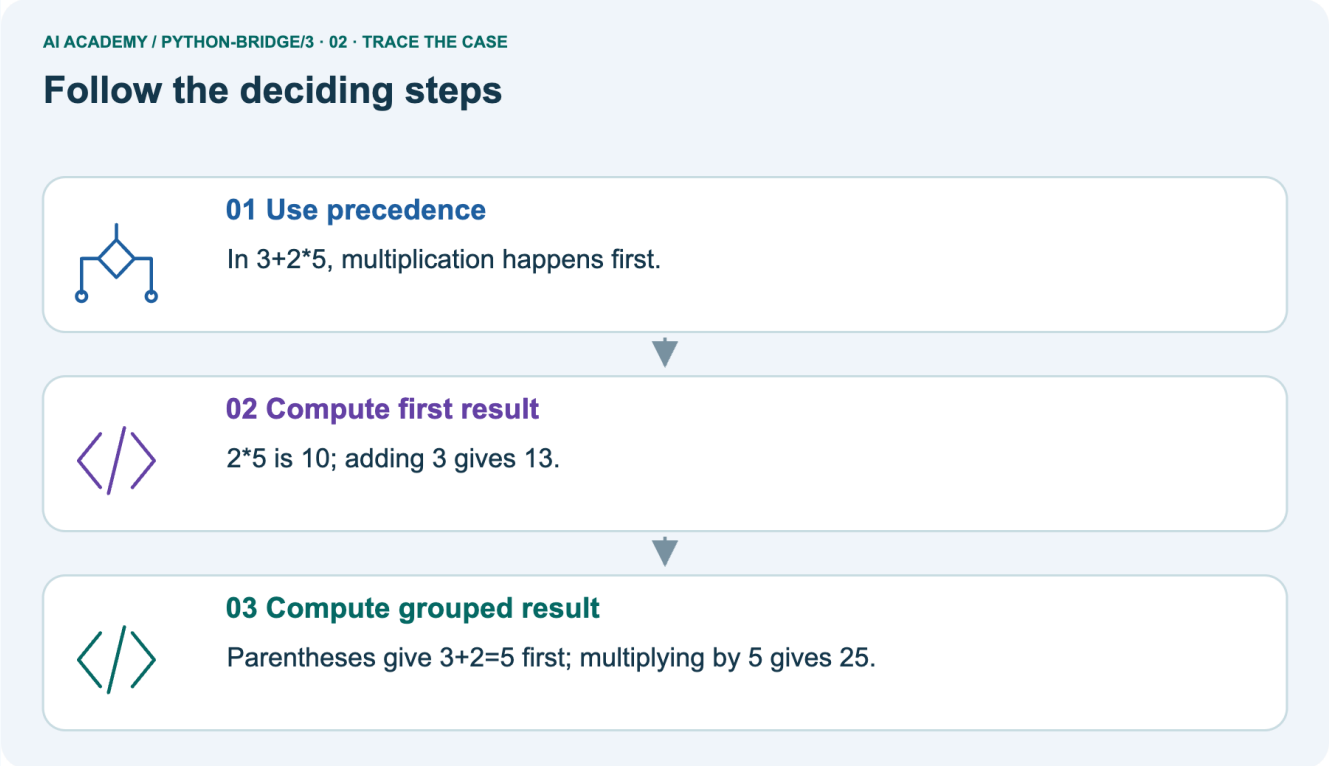
Follow the labelled picture with the matching explanation. Open the vector diagram

Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group. Supplied facts: Python 3: print(3+2*5); print((3+2)*5).

Inputs are dimensionless for this arithmetic exercise.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

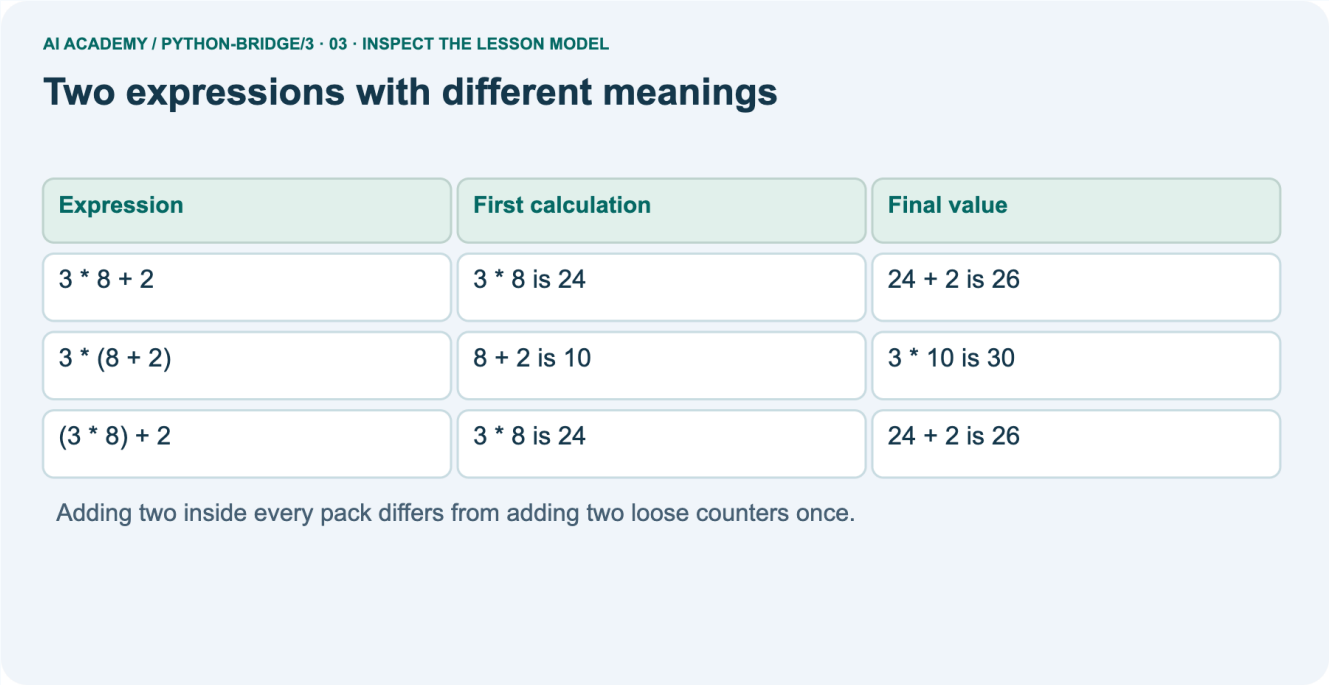


Follow the labelled picture with the matching explanation. Open the vector diagram

Use **precedence**: In 3+2*5, multiplication happens first. → Compute first result: 2*5 is 10; adding 3 gives 13. → Compute grouped result: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Two expressions with different meanings

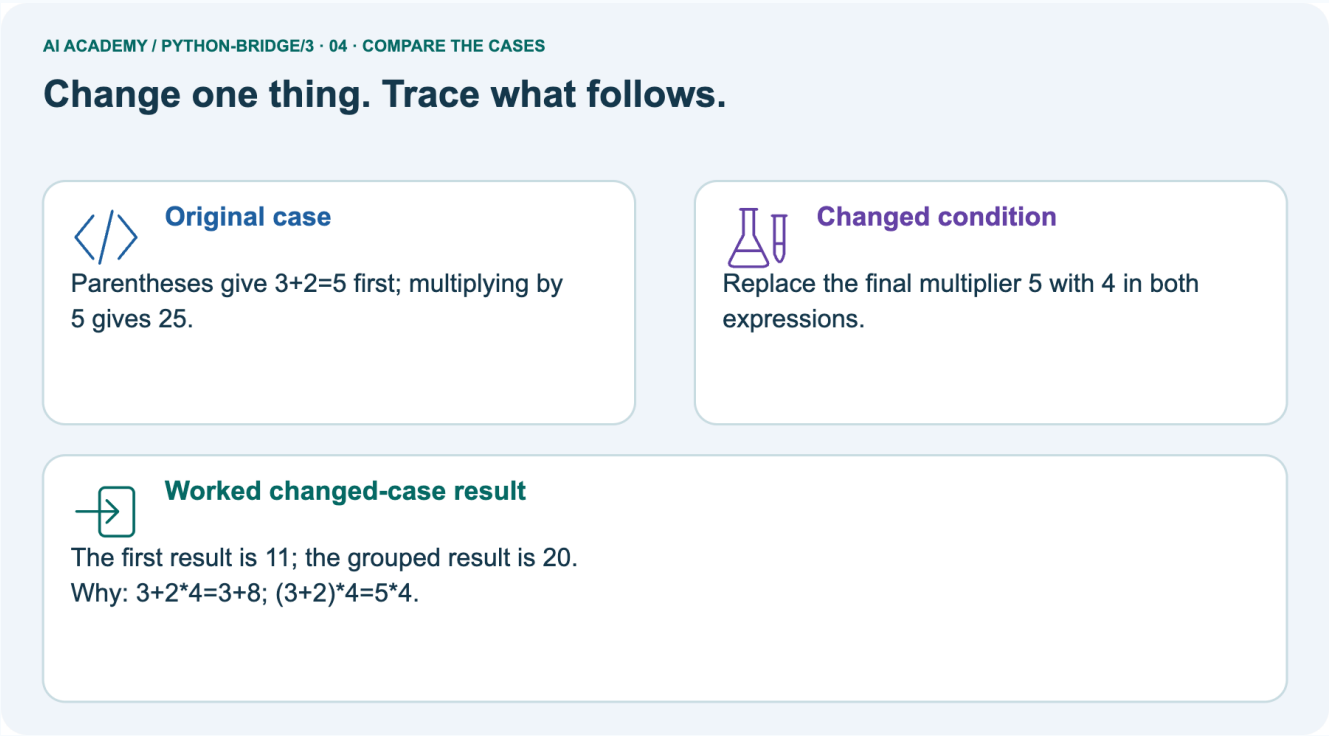


Follow the labelled picture with the matching explanation. Open the vector diagram

Two **expression**s with different meanings. 3 * 8 + 2 | 3 * 8 is 24 | 24 + 2 is 26. 3 * (8 + 2) | 8 + 2 is 10 | 3 * 10 is 30. (3 * 8) + 2 | 3 * 8 is 24 | 24 + 2 is 26

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



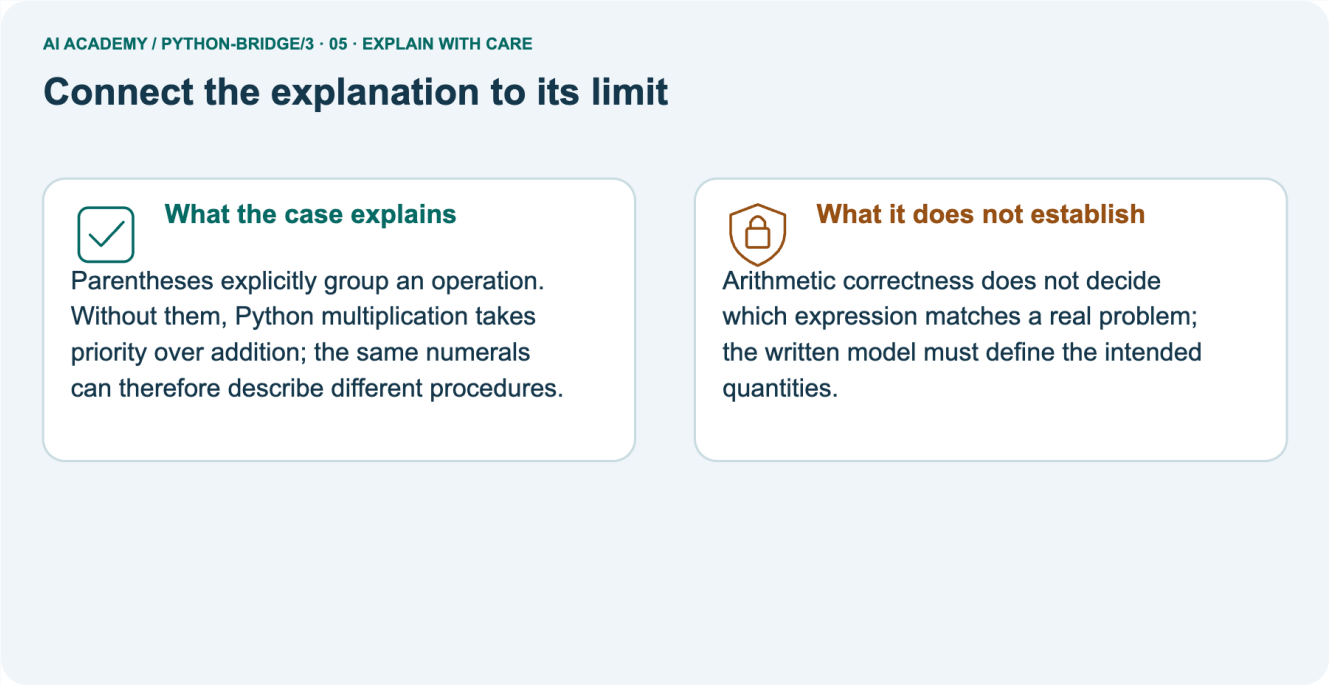
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. Changed condition: Replace the final multiplier 5 with 4 in both **expression**s. Expected result: The first result is 11; the grouped result is 20.

Reason: 3+2*4=3+8; (3+2)*4=5*4.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



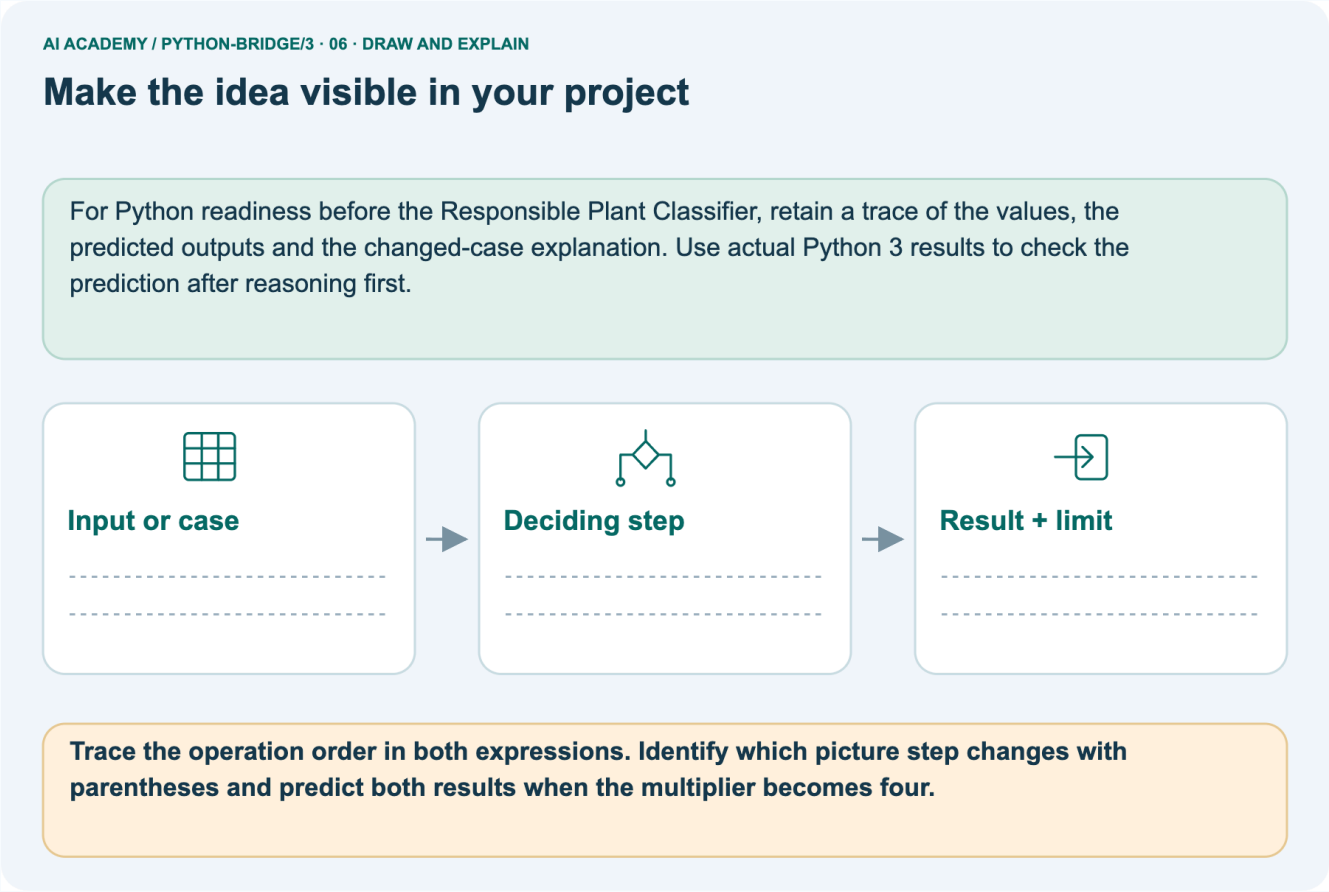
Follow the labelled picture with the matching explanation. Open the vector diagram

Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Limit: Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Trace the operation order in both **expression**s.

Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.